

EAGLE RIVER, WI, USA

WMO: 726504

Lat: 45.932N

Lon: 89.269W

Elev: 1642

StdP: 13.84

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-10.6	-23.8	1.6	-15.3	-17.6	2.2	-8.6	20.4	18.0	18.6	16.5	2.3	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	84.1	68.9	81.7	67.3	79.4	65.6	73.1	79.3	71.0	77.3	69.1	75.2	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	124.6	76.5	69.6	115.7	74.3	66.5	103.6	71.7	37.8	79.6	35.9	77.7	34.2	75.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	17.0	15.3	-24.4	89.3	3.4	2.9	-26.9	91.4	-28.9	93.2	-30.8	94.8	-33.2	96.9		
			-24.8	75.8	3.4	2.9	-27.2	77.9	-29.2	79.5	-31.0	81.2	-33.5	83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	13.4	15.5	26.2	39.6	52.2	61.4	65.6	63.1	56.3	43.6	30.9	18.6	(6)
	DBStd	20.89	12.40	12.32	12.51	9.69	8.99	7.15	5.96	6.12	8.38	8.76	9.70	11.59	(7)
	HDD50	5083	1134	966	743	331	81	5	0	0	32	243	574	974	(8)
	HDD65	9089	1599	1386	1203	762	407	151	66	109	280	665	1022	1439	(9)
	CDD50	1685	0	0	6	20	150	346	485	408	223	45	2	0	(10)
	CDD65	213	0	0	0	1	11	42	86	51	21	1	0	0	(11)
	CDH74	2206	0	0	1	17	172	451	889	470	195	11	0	0	(12)
	CDH80	422	0	0	0	1	35	83	192	81	30	0	0	0	(13)
	WSAvg	6.0	6.3	6.4	6.3	7.0	6.7	5.6	5.1	4.8	5.4	6.1	6.4	6.2	(14)
	PrecAvg	31.5	1.2	0.9	1.8	2.6	3.6	3.9	4.3	3.2	3.7	3.3	1.9	1.3	(15)
(16) Precipitation	PrecMax	40.7	2.9	2.0	4.3	5.7	7.4	7.9	8.1	6.2	8.2	6.3	5.0	2.5	(16)
	PrecMin	24.3	0.3	0.0	0.4	0.7	1.2	2.0	1.4	1.5	0.8	1.1	0.4	0.3	(17)
	PrecStd	4.1	0.7	0.5	1.0	1.3	1.5	1.4	1.7	1.3	1.7	1.5	1.0	0.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	49.8	67.8	76.5	84.1	86.6	88.4	86.3	83.8	75.1	61.5	46.0	(19)
		MCWB	37.6	41.8	56.6	58.6	65.2	69.8	72.8	69.8	69.9	62.0	52.1	41.8	(20)
	2%	DB	36.7	42.8	56.9	68.3	79.2	82.0	84.4	81.9	79.4	69.9	54.8	39.7	(21)
		MCWB	34.1	37.8	47.8	52.1	61.2	67.1	69.6	68.2	66.1	60.5	48.8	36.2	(22)
	5%	DB	33.7	37.0	50.4	63.2	73.5	79.3	81.8	79.2	75.2	63.9	50.0	36.2	(23)
		MCWB	32.5	33.8	44.1	48.4	57.8	64.9	68.0	66.5	64.1	56.7	44.3	34.0	(24)
	10%	DB	30.1	32.6	44.6	57.3	69.7	75.5	79.4	75.9	71.9	59.1	45.4	33.7	(25)
		MCWB	28.9	30.3	38.8	45.7	56.5	62.8	66.2	64.8	62.8	52.1	41.4	32.4	(26)
	0.4%	WB	37.3	43.6	58.3	61.2	70.1	74.1	76.9	74.5	72.0	69.5	54.5	43.5	(27)
		MCDB	40.0	46.1	64.5	70.1	78.4	82.0	84.6	79.8	78.2	71.6	57.8	45.3	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.4	38.0	51.6	55.1	65.6	71.0	73.8	71.8	69.4	62.5	50.4	37.3	(29)
		MCDB	36.4	41.6	54.0	64.4	72.8	78.2	80.1	77.2	75.1	65.6	54.5	38.9	(30)
	5%	WB	32.2	34.3	44.2	50.5	62.4	68.5	71.4	69.6	67.1	57.4	45.1	34.2	(31)
		MCDB	33.6	37.3	49.2	60.4	68.8	75.2	77.8	75.2	72.5	63.2	48.2	35.6	(32)
	10%	WB	28.9	30.4	38.7	46.7	59.0	66.1	69.1	67.7	64.5	53.5	41.5	32.1	(33)
		MCDB	29.9	32.2	43.5	55.7	66.6	72.2	75.4	74.0	69.1	58.3	45.1	33.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	19.7	21.0	21.7	23.7	22.3	23.1	22.3	21.9	18.1	14.2	14.3	(35)
		MCDBR	17.6	21.2	24.9	32.8	29.4	25.7	25.2	24.7	25.7	25.8	22.7	16.8	(36)
		MCWBR	15.1	16.0	15.9	17.8	13.7	12.5	12.2	12.4	13.5	16.0	15.8	13.7	(37)
	5% WB	MCDBR	16.6	19.1	22.4	27.6	22.6	21.8	21.4	20.3	20.2	21.8	19.8	14.0	(38)
		MCWBR	14.5	14.5	15.6	17.4	12.9	12.6	12.3	12.0	12.5	15.8	15.4	12.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.239	0.266	0.309	0.332	0.363	0.363	0.362	0.360	0.342	0.306	0.279	0.249	(40)
	taud		2.396	2.311	2.344	2.430	2.388	2.406	2.431	2.436	2.488	2.578	2.546	2.442	(41)
	Ebn at Noon		283	292	292	294	287	286	285	281	279	276	265	266	(42)
	Edh at Noon		22	30	34	34	37	36	35	34	29	23	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		449	782	1180	1447	1708	1843	1906	1607	1238	757	439	319	(44)
	RadStd		43	85	112	145	142	145	96	102	78	82	49	36	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (36 neighbors)	N/A	N/A	N/A	N/A	-1.78	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page